

DESIGN AND IMPLEMENTATION OF A ZONE-BASED BLE INDOOR LOCALIZATION SYSTEM USING ESP32

A Project Report

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in partial fulfillment for the award of the degree

Of

BACHELOR OF ENGINEERING

In

ELECTRONICS AND COMMUNICATION ENGINEERING



CHENNAI INSTITUTE OF TECHNOLOGY

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MARCH 2026

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ACKNOWLEDGEMENT

We express our gratitude to our Chairman **Shri. P. SRIRAM** and all trust members of Chennai Institute of Technology for providing the facility and opportunity to do this project as a part of our undergraduate course.

We are grateful to our Principal **Dr. A. RAMESH, M.E., Ph.D.** for providing us the facility and encouragement during the course of our work.

We sincerely thank our Head of the Department, **Dr. T.S.ARULANANTH, M.Tech., (Ph.D).**, Department of Electronics and Communication Engineering for having provided us valuable guidance, resources and timely suggestions throughout our work.

We sincerely thank our Project Guide and Supervisor, **Ms. M. HARSHANYA, M.E., (Ph.D).**, Assistant Professor, Department of Electronics and Communication Engineering for having provided us valuable guidance, resources and timely suggestions throughout our work.

We would like to extend our thanks to our **Faculty coordinators of the Department of Electronics And Communication Engineering**, for their valuable suggestions throughout this project.

We wish to extend our sincere thanks to all **Faculty members of the Department of Electronics And Communication Engineering** for their valuable suggestions and their kind cooperation for the successful completion of our project.

We wish to acknowledge the help received from the **Lab Instructors of the Department of Electronics And Communication Engineering** and others for providing valuable suggestions and for the successful completion of the project.

PREFACE

We, **Akash M (22EC009)** and **Siddarth S (22EC305)**, students of the Department of Electronics and Communication Engineering, hereby present the project report titled **“Design and Implementation of a Zone-Based BLE Indoor Localization System Using ESP32”** as a part of our final year academic curriculum.

Indoor positioning has become essential in smart environments where GPS signals fail due to attenuation and multipath effects. This project addresses the challenge by developing a low-cost and low-power indoor localization system using Bluetooth Low Energy (BLE) beacons and the ESP32 platform. Unlike conventional coordinate-based systems, the proposed solution focuses on zone-level localization, providing stable and reliable proximity detection without complex calibration or high computational requirements.

The system performs real-time RSSI acquisition, lightweight sliding window filtering, and hysteresis-based zone stabilization directly on the embedded device. Experimental validation demonstrates improved signal stability and consistent zone detection accuracy under practical indoor conditions.

This work emphasizes a fully embedded, edge-based architecture where all processing and decision-making are executed locally on the ESP32 without reliance on cloud infrastructure. The proposed approach enhances scalability, reduces latency, and makes the system suitable for resource-constrained indoor applications such as navigation, asset tracking, and proximity-based services.

Through this project, we strengthened our understanding of wireless communication principles, embedded system design, and real-time signal processing while translating theoretical knowledge into a functional and validated prototype.



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ABSTRACT

Indoor localization has become an essential component of smart infrastructure applications where Global Positioning System (GPS) signals are unreliable. This project presents the design and implementation of a zone-based Bluetooth Low Energy (BLE) indoor localization system using ESP32 development boards. The system employs fixed BLE beacon nodes and an ESP32-based receiver operating in continuous scanning mode to acquire Received Signal Strength Indicator (RSSI) values. To improve signal stability, a sliding window averaging technique is implemented to reduce short-term RSSI fluctuations caused by multipath propagation and environmental interference. A threshold-based classification algorithm with hysteresis stabilization is applied to determine user presence within predefined indoor zones. Additionally, a log-distance path loss model is utilized to estimate approximate distance ranges. Experimental evaluation demonstrates improved RSSI stability and classification accuracy after filtering, achieving reliable zone-level localization suitable for applications such as asset tracking, indoor navigation, and smart building monitoring. The system operates entirely on embedded hardware without cloud dependency, ensuring low latency and energy-efficient performance. The proposed architecture emphasizes low hardware complexity and cost-effective deployment using readily available ESP32 modules. The results validate that zone-level localization provides a practical and scalable solution for real-time indoor positioning in dynamic environments.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

BLE	Bluetooth Low Energy
RSSI	Received Signal Strength Indicator
ESP32	Espressif 32-bit Microcontroller
UWB	Ultra-Wideband
Wi-Fi	Wireless Fidelity
dBm	Decibel-milliwatts
P_o	Reference RSSI at 1 meter
d	Estimated distance
d_o	Reference distance
n	Path loss exponent
ToF	Time of Flight
TDoA	Time Difference of Arrival

CHAPTER 1

INTRODUCTION

1.1 GENERAL INTRODUCTION

Indoor localization has emerged as a critical enabling technology in smart infrastructure, healthcare monitoring, asset tracking, and industrial automation. While the Global Positioning System (GPS) provides reliable positioning in outdoor environments, it fails to deliver accurate location information indoors due to signal attenuation, multipath propagation, and interference caused by walls, ceilings, and other structural components. This limitation has created a demand for alternative indoor positioning technologies that are low-cost, energy-efficient, and scalable.

Several technologies such as Wi-Fi fingerprinting, Ultra-Wideband (UWB), RFID, and Bluetooth Low Energy (BLE) have been explored for indoor localization. Among these, BLE has gained significant attention due to its low power consumption, affordability, and ease of deployment. BLE beacons continuously broadcast signals that can be detected by nearby receivers, enabling proximity detection and location estimation based on Received Signal Strength Indicator (RSSI) values.

However, conventional RSSI-based distance estimation methods often suffer from signal fluctuations caused by environmental factors such as obstacles, human movement, and multipath effects. Instead of focusing on precise coordinate-level localization, a zone-based approach provides a more stable and practical solution for real-world indoor applications. Zone-based positioning divides an indoor environment into predefined regions and

determines the user's presence within a specific zone rather than calculating exact coordinates.

This project focuses on the design and implementation of a zone-based Bluetooth Low Energy (BLE) Indoor Positioning System using ESP32 modules and BLE beacons. The proposed system utilizes RSSI measurements to classify indoor zones efficiently while maintaining low hardware cost and reduced power consumption. The system is designed to support applications in hospitals, educational institutions, shopping complexes, and office environments where zone-level localization is sufficient and practical.

1.2 PROBLEM STATEMENT

Indoor positioning remains a challenging problem due to the limitations of traditional localization technologies. The Global Positioning System (GPS), although highly effective in outdoor environments, fails indoors because satellite signals are significantly attenuated by building structures such as concrete walls, ceilings, and metal reinforcements. This makes GPS unsuitable for accurate indoor navigation and tracking applications.

Existing indoor positioning technologies such as Wi-Fi fingerprinting, Ultra-Wideband (UWB), and RFID-based systems offer varying levels of accuracy; however, they introduce challenges related to high infrastructure cost, complex deployment, increased power consumption, or the need for extensive calibration. In particular, precise coordinate-based localization systems often require multiple anchors, advanced signal processing algorithms, and continuous recalibration to maintain accuracy. Bluetooth Low Energy (BLE)-based systems offer a low-cost and energy-efficient alternative.

These fluctuations make precise distance calculation unreliable in dynamic indoor environments.

Therefore, there is a need for a practical indoor positioning solution that:

- Minimizes hardware complexity,
- Reduces deployment cost,
- Consumes low power,
- Maintains stable performance in dynamic environments.

A zone-based BLE indoor positioning system addresses these challenges by classifying user presence into predefined indoor zones instead of attempting exact coordinate estimation. This approach improves stability, reduces computational complexity, and provides sufficient accuracy for real-world applications such as hospital navigation, classroom monitoring, asset tracking, and smart building management.

1.3 OBJECTIVES OF THE PROJECT

The main objective of this project is to design and implement a zone-based Bluetooth Low Energy (BLE) Indoor Positioning System using ESP32 modules and BLE beacons.

The specific objectives of the project are:

- To design a low-cost indoor positioning architecture using BLE beacons and ESP32 modules.
- To collect and analyse RSSI (Received Signal Strength Indicator) values for indoor signal strength evaluation.

- To implement a zone classification algorithm based on predefined RSSI threshold values.
- To minimize hardware complexity by reducing the number of required beacons.
- To optimize power consumption for efficient long-term deployment.
- To evaluate system performance in terms of zone detection accuracy and stability.
- To develop a scalable solution suitable for applications in hospitals, educational institutions, offices, and smart buildings.

1.4 SCOPE OF THE PROJECT

The scope of this project is limited to the design and implementation of a zone-based indoor positioning system using Bluetooth Low Energy (BLE) technology. The system focuses on detecting user presence within predefined indoor zones rather than providing precise coordinate-level localization.

The project includes the deployment of BLE beacons in selected indoor areas and the use of ESP32 modules for scanning and processing RSSI values. Based on the collected signal strength data, the system classifies the indoor environment into predefined zones using threshold-based decision logic.

The scope of the system covers:

- Indoor environments such as classrooms, laboratories, hospital wards, office rooms, and shopping areas.
- Zone-level localization within a limited indoor range.
- Low-cost and low-power deployment scenarios.

- Basic performance evaluation under static and moderately dynamic conditions.

The system does not aim to provide high-precision coordinate estimation, centimeter-level accuracy, or large-scale multi-floor navigation. Advanced machine learning-based localization techniques and complex trilateration algorithms are beyond the current scope of this project.

1.5 JUSTIFICATION OF THE PROJECT

Indoor positioning has become increasingly important for modern smart infrastructure applications where reliable location information is required inside buildings. Conventional positioning technologies such as the Global Positioning System (GPS) cannot provide accurate indoor localization due to signal attenuation caused by walls, ceilings, and other structural obstacles. Although several indoor positioning technologies such as Wi-Fi fingerprinting and Ultra-Wideband (UWB) have been developed, they often require expensive infrastructure, complex calibration procedures, or high computational resources.

Bluetooth Low Energy (BLE) technology provides a practical alternative for indoor localization due to its low power consumption, low deployment cost, and ease of implementation using widely available embedded platforms such as ESP32. However, conventional RSSI-based localization methods suffer from signal fluctuations caused by multipath propagation and environmental interference. Therefore, a simplified zone-based localization approach combined with lightweight signal filtering techniques can provide stable and reliable indoor positioning without requiring complex algorithms or expensive hardware.

The proposed system addresses these challenges by implementing a low-cost, zone-based BLE indoor localization system using ESP32 modules. The system performs RSSI filtering and zone classification directly on the embedded device, reducing computational overhead and eliminating the need for cloud-based processing. This approach provides a practical and scalable solution suitable for applications such as asset tracking, indoor navigation, and smart building monitoring.

1.6 ORGANIZATION OF THE REPORT

This report is organized into seven chapters.

Chapter 1 presents the introduction, problem statement, objectives, scope, and structure of the project.

Chapter 2 reviews the existing literature related to indoor positioning systems, BLE-based localization techniques, and RSSI-based distance estimation methods.

Chapter 3 describes the overall system architecture, including hardware design and system workflow.

Chapter 4 explains the hardware implementation, including ESP32 configuration and BLE beacon deployment.

Chapter 5 discusses the software implementation, including RSSI processing and zone classification logic.

Chapter 6 presents the experimental results and performance analysis of the proposed system.

Chapter 7 concludes the project and outlines possible future enhancements.

CHAPTER 2

LITERATURE REVIEW

2.1 OVERVIEW OF INDOOR LOCALIZATION SYSTEMS

Indoor localization has been an active research domain for more than two decades due to the limitations of GPS in enclosed environments. Early research efforts focused on radio frequency-based positioning techniques such as RADAR, which utilized Wi-Fi signal strength measurements for in-building localization. Bahl and Padmanabhan (2000) introduced one of the earliest RSSI-based indoor localization systems, demonstrating the feasibility of RF-based indoor tracking.

Subsequent surveys by Hightower and Borriello (2001) and Gu et al. (2009) provided comprehensive overviews of wireless indoor positioning technologies, categorizing them into proximity-based, fingerprinting-based, and trilateration-based approaches. More recent survey studies by Zafari et al. (2023) and Liu et al. (2022) have analysed modern indoor positioning technologies including Wi-Fi, Ultra-Wideband (UWB), RFID, BLE, and hybrid systems, highlighting trade-offs in accuracy, cost, power consumption, and deployment complexity.

Indoor positioning systems are generally evaluated based on accuracy, scalability, infrastructure requirements, computational complexity, and robustness to environmental dynamics. While high-precision systems aim to estimate exact coordinates, practical applications such as asset tracking, indoor navigation, and smart building services often require only region-level or zone-level localization. This distinction has motivated research toward simplified and robust indoor positioning architectures.

2.2 WI-FI BASED INDOOR LOCALIZATION

Wi-Fi-based indoor localization systems are among the earliest and most widely researched indoor positioning approaches due to the widespread availability of wireless infrastructure in buildings. These systems primarily utilize Received Signal Strength Indicator (RSSI), Channel State Information (CSI), or time-based measurements to estimate user location.

One of the pioneering systems in this domain was the RADAR system proposed by Bahl and Padmanabhan (2000), which demonstrated indoor positioning using RSSI measurements from Wi-Fi access points. This approach introduced the concept of fingerprinting, where signal strength measurements are collected during an offline training phase and later matched during real-time localization.

Fingerprinting-based Wi-Fi localization techniques can achieve moderate to high accuracy under controlled conditions. However, they require extensive site surveys, large training datasets, and frequent recalibration to maintain performance in dynamic indoor environments. Environmental changes such as furniture rearrangement, human movement, and interference significantly affect signal patterns, increasing maintenance complexity.

Channel State Information (CSI)-based techniques have also been explored to enhance localization accuracy. While CSI provides finer-grained signal characteristics compared to RSSI, it requires specialized hardware support and higher computational resources.

Despite their feasibility, Wi-Fi-based systems suffer from several limitations, including high power consumption, infrastructure dependency, and sensitivity to environmental variations. These drawbacks make Wi-Fi localization less suitable for low-power embedded systems and resource-constrained

applications. As a result, research has gradually shifted toward alternative technologies such as Bluetooth Low Energy (BLE), which offers lower power consumption and simplified deployment.

2.3 ULTRA-WIDEBAND (UWB) BASED LOCALIZATION

Ultra-Wideband (UWB) is a high-precision indoor localization technology that operates by transmitting very short-duration pulses over a wide frequency spectrum. Unlike RSSI-based systems, UWB primarily relies on Time-of-Flight (ToF) or Time-Difference-of-Arrival (TDoA) measurements to estimate distance between anchors and mobile devices. These time-based techniques provide significantly higher localization accuracy compared to conventional signal strength-based approaches.

Several studies have demonstrated that UWB systems can achieve centimetre-level positioning accuracy under controlled indoor conditions. This high precision makes UWB suitable for applications such as industrial automation, robotics, and real-time asset tracking in warehouses.

However, UWB systems require specialized hardware modules and synchronized anchor nodes, which increase deployment complexity and infrastructure cost. Additionally, large-scale deployment demands careful calibration and precise anchor placement to maintain accuracy. The need for dedicated hardware also limits integration with low-cost embedded systems.

Although UWB provides superior accuracy compared to Wi-Fi and BLE-based RSSI systems, its high implementation cost and hardware requirements make it less practical for low-budget and resource-constrained indoor environments. Consequently, research interest has grown toward more affordable and energy-efficient technologies such as Bluetooth Low Energy

(BLE), particularly for zone-level indoor localization where centimetre-level accuracy is not required.

2.4 BLE-BASED INDOOR LOCALIZATION

Bluetooth Low Energy (BLE) has emerged as a promising technology for indoor localization due to its low power consumption, low hardware cost, and widespread availability in embedded platforms and consumer devices. BLE beacons periodically broadcast advertising packets containing unique identifiers, which can be detected by nearby receivers to estimate proximity or location.

Most BLE-based indoor positioning systems rely on Received Signal Strength Indicator (RSSI) measurements to infer distance between a beacon and a receiver. Two major localization approaches are commonly adopted in BLE systems: fingerprinting-based localization and trilateration-based localization.

Fingerprinting methods collect RSSI signatures during an offline training phase and match real-time measurements with stored signal maps. While this technique can achieve reasonable accuracy, it requires extensive data collection, dense beacon deployment, and frequent recalibration to adapt to environmental changes. These requirements increase maintenance overhead and reduce scalability.

Trilateration-based approaches estimate distance from RSSI values using path loss models and calculate user position geometrically. Although trilateration eliminates the need for an offline training phase, it is highly sensitive to RSSI fluctuations caused by multipath propagation, shadowing, human movement, and environmental interference. As a result, coordinate-level accuracy is often unstable in dynamic indoor environments.

Recent research has focused on improving BLE localization performance through lightweight signal processing techniques such as moving average filtering, median filtering, Kalman filtering, and adaptive smoothing. While advanced filtering techniques can enhance signal stability, they may increase computational complexity and are sometimes implemented on centralized servers rather than on low-power embedded devices.

Furthermore, many BLE-based systems aim to estimate precise spatial coordinates, which increases algorithmic complexity and infrastructure requirements. In practical indoor applications such as aisle-level navigation, room-level tracking, and section-based monitoring, zone-level localization is often sufficient and more robust to signal instability.

Therefore, BLE technology provides a suitable balance between cost, power efficiency, scalability, and practical accuracy, especially when combined with simplified zone-based localization strategies that prioritize stability over centimetre-level precision.

2.5 RSSI FILTERING AND STABILITY ENHANCEMENT TECHNIQUES

One of the primary challenges in BLE-based indoor localization is the instability of Received Signal Strength Indicator (RSSI) measurements. Indoor radio propagation is significantly affected by multipath fading, reflection from walls and furniture, shadowing, non-line-of-sight conditions, and human movement. These factors cause rapid fluctuations in RSSI values, leading to inaccurate distance estimation and unstable localization results.

To mitigate RSSI instability, various signal processing techniques have been proposed in the literature. Simple statistical methods such as moving average filtering and median filtering are commonly used to reduce short-term signal

variations. These techniques are computationally lightweight and suitable for real-time embedded implementations.

More advanced techniques such as Kalman filtering and adaptive filtering models have also been explored to enhance signal stability. Kalman filters can effectively model dynamic systems and reduce measurement noise; however, they introduce additional computational complexity and require parameter tuning. In many implementations, such filtering methods are executed on centralized servers rather than directly on resource-constrained embedded devices.

Recent research trends emphasize lightweight edge-based processing, where signal conditioning and localization decisions are performed directly on embedded platforms. This approach reduces communication overhead, minimizes latency, and improves scalability in large indoor deployments. However, achieving stable localization using only low-complexity algorithms remains a challenge.

In addition to filtering techniques, hysteresis-based decision mechanisms have been introduced to prevent frequent zone switching when RSSI values fluctuate around threshold boundaries. Therefore, combining lightweight RSSI averaging techniques with hysteresis-based zone locking provides a practical solution for achieving stable and reliable zone-level indoor localization on low-power embedded systems such as the ESP32.

2.6 RESEARCH GAP

A review of existing indoor localization systems indicates that significant research has been conducted on coordinate-level positioning using Wi-Fi, UWB, RFID, and BLE technologies. While high-precision systems can achieve impressive accuracy under controlled conditions, they often require

dense infrastructure, extensive calibration, complex signal processing algorithms, and centralized computation.

Wi-Fi-based fingerprinting systems demand labor-intensive site surveys and frequent recalibration. UWB-based systems provide centimeter-level accuracy but rely on specialized hardware and incur high deployment costs. BLE-based trilateration and fingerprinting methods, although more affordable, remain highly sensitive to RSSI fluctuations and environmental dynamics.

Furthermore, many existing BLE localization solutions focus primarily on precise coordinate estimation rather than practical zone-level classification. Coordinate-based approaches increase computational complexity and infrastructure requirements, which may not be necessary for applications such as room-level detection, aisle-level navigation, or section-based monitoring.

Another limitation observed in prior work is the reliance on server-side or cloud-based processing for signal filtering and localization decisions. Such architectures increase communication overhead, latency, and power consumption, making them less suitable for fully embedded and resource-constrained deployments.

Although lightweight RSSI filtering techniques have been proposed, there remains a need for a simple, calibration-free, low-cost, and fully embedded zone-based localization system that operates reliably in dynamic indoor environments without complex modelling or dense beacon networks.

Therefore, this project addresses the identified research gap by proposing and implementing a zone-based BLE indoor localization system using ESP32, where signal processing, RSSI filtering, and zone decision logic are executed

entirely on the embedded platform with minimal infrastructure and computational complexity.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 SYSTEM ARCHITECTURE OVERVIEW

The proposed system is a zone-based Bluetooth Low Energy (BLE) indoor localization architecture designed to operate entirely on a low-power embedded platform. The system utilizes fixed BLE beacon nodes deployed at known indoor locations and an ESP32 module functioning as the central processing and localization unit.

The overall system is designed to achieve reliable zone-level localization rather than precise coordinate estimation. The localization process is performed through continuous BLE scanning, RSSI acquisition, lightweight signal filtering, and threshold-based zone classification.

The system architecture consists of the following main components:

- BLE Beacon Nodes – Fixed transmitters placed at predefined indoor locations that periodically broadcast BLE advertisement packets containing unique identifiers.
- ESP32 Receiver Module – Operates in continuous scanning mode to receive advertisement packets and extract RSSI values in real time.
- RSSI Processing Unit – Implements sliding window averaging to reduce short-term signal fluctuations.
- Zone Decision Logic – Classifies the user's location into predefined proximity zones based on filtered RSSI thresholds.

- Hysteresis-Based Zone Locking Mechanism – Prevents frequent zone switching when RSSI values fluctuate near decision boundaries.

All signal processing, filtering, and decision-making operations are executed locally on the ESP32 platform without reliance on external servers or cloud-based computation. This fully embedded architecture reduces latency, minimizes communication overhead, and enhances scalability in resource-constrained indoor environments.

The system is designed to be low-cost, energy-efficient, and easily deployable in indoor environments such as classrooms, laboratories, hospital wards, office spaces, and retail stores.

3.2 BLOCK DIAGRAM AND SYSTEM COMPONENTS

The overall architecture of the proposed zone-based BLE indoor localization system is illustrated in Figure 3.1. The block diagram presents the sequential flow of signal acquisition, processing, and decision-making implemented on the ESP32 platform. The system integrates BLE communication, RSSI signal conditioning, zone classification, and output generation within a fully embedded framework to achieve stable and reliable zone-level indoor localization.

The architecture follows a pipeline-based processing model, where each functional block performs a specific operation before passing the processed data to the next stage. This structured design ensures modularity, reduced computational complexity, and efficient real-time execution. The figure highlights how raw beacon signals are transformed into meaningful zone-level outputs through lightweight embedded processing.

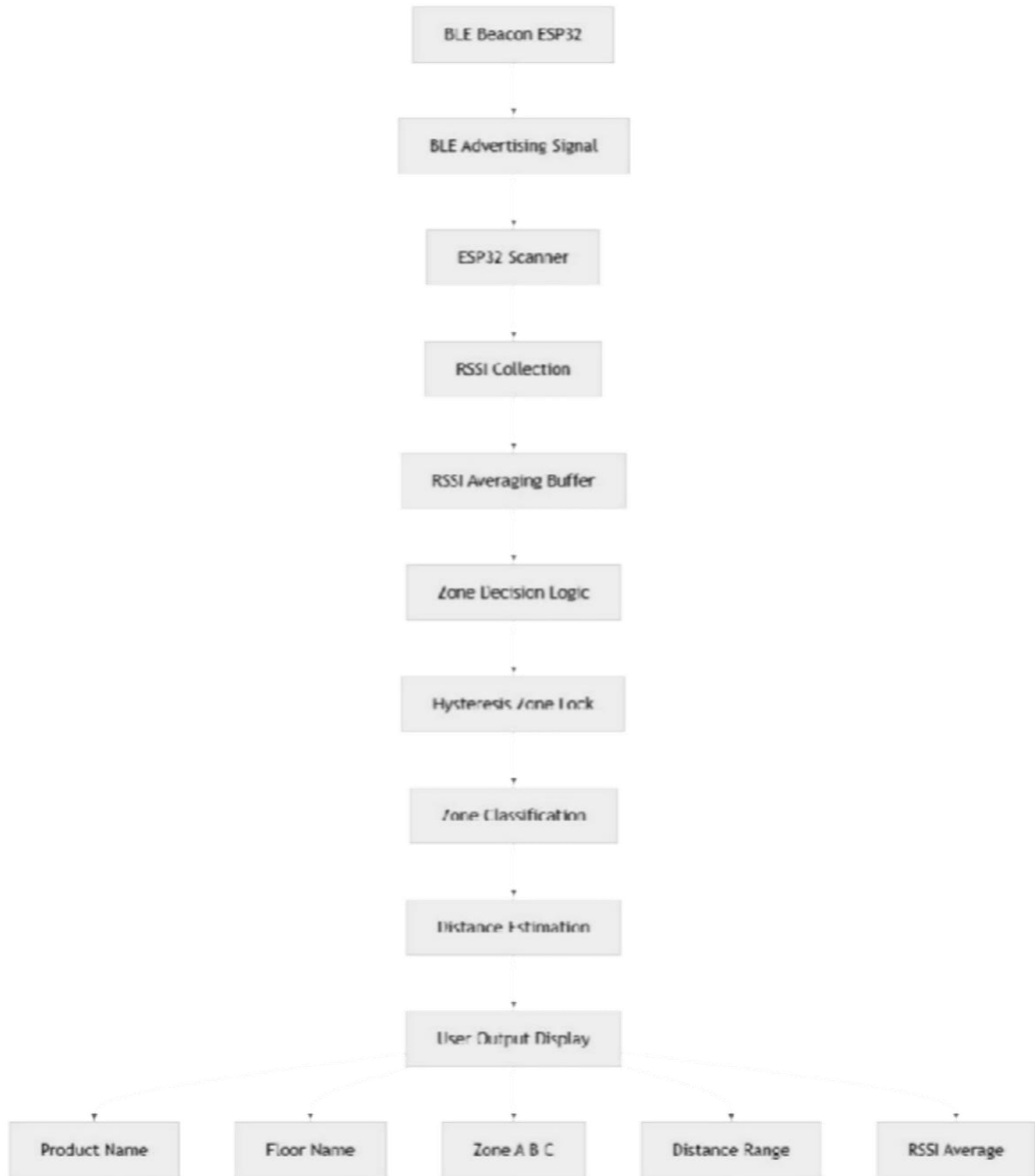


Figure 3.1 Block Diagram of the Proposed Zone-Based BLE Indoor Localization System

3.3 SYSTEM WORKFLOW

The operational workflow of the proposed zone-based BLE indoor localization system follows a sequential pipeline executed entirely on the ESP32 platform. The workflow consists of the following stages:

Step 1: BLE Advertisement Broadcasting

Fixed BLE beacon nodes periodically transmit advertisement packets containing unique identifiers.

Step 2: Continuous BLE Scanning

The ESP32 operates in continuous scanning mode and receives advertisement packets from nearby beacons.

Step 3: RSSI Acquisition

Upon receiving each advertisement packet, the ESP32 extracts the corresponding RSSI value associated with the detected beacon.

Step 4: Sliding Window RSSI Averaging

A predefined number of recent RSSI samples are stored in a buffer. The average RSSI value is computed to reduce short-term fluctuations and suppress signal noise.

The filtered RSSI value using sliding window averaging is computed as:

$$\bar{r} = \frac{1}{N} \sum_{i=1}^N r(i)$$

Where:

$r(i)$ = Instantaneous RSSI sample

N = Window size

$\bar{r}$ = Averaged RSSI value

This averaging process reduces short-term signal fluctuations and improves stability for zone classification.

Step 5: Threshold-Based Zone Decision

The filtered RSSI value is compared with predefined threshold levels corresponding to Near, Mid, and Far zones.

Step 6: Hysteresis-Based Zone Locking

To prevent frequent switching between adjacent zones, a hysteresis mechanism validates zone transitions only if the new condition persists across multiple measurement cycles.

Step 7: Distance Range Mapping and Output Generation

Once the zone is confirmed, an approximate distance range is assigned. The system outputs the zone classification, filtered RSSI value, estimated distance range, and contextual information.

The approximate distance between the beacon and receiver is estimated using the log-distance path loss model:

$$d = d_0 \times 10^{\frac{RSSI_0 - RSSI}{10 \times \alpha}}$$

Where:

d = Estimated distance

d_0 = Reference distance

$RSSI_0$ = RSSI at reference distance

$RSSI$ = Filtered RSSI value

α = Path loss exponent

The calculated distance is not used for coordinate estimation but for assigning predefined proximity zones.

All processing stages are executed locally on the ESP32 without reliance on cloud-based computation, ensuring low latency and energy-efficient operation.



Figure 3.2 the overall decision-making process of the proposed system.

3.4 DESIGN CONSIDERATIONS

The proposed zone-based BLE indoor localization system was designed with a focus on simplicity, stability, and practical deployment in resource-constrained environments. Several architectural decisions were made to balance accuracy, computational complexity, power consumption, and infrastructure requirements.

Zone-Level Localization Instead of Coordinate Estimation:

Rather than estimating precise spatial coordinates, the system performs zone-based classification. Coordinate-level localization using trilateration or fingerprinting increases computational complexity and sensitivity to RSSI fluctuations. For applications such as room-level tracking, aisle detection, and section-based navigation, zone-level accuracy is sufficient and more robust.

Lightweight Sliding Window Averaging:

Advanced filtering techniques such as Kalman filtering can improve RSSI stability but introduce higher computational overhead and parameter tuning complexity. A sliding window moving average was selected due to its simplicity, low memory requirement, and suitability for real-time execution on the ESP32 platform.

Hysteresis-Based Zone Locking:

To prevent frequent zone oscillation near decision boundaries, a hysteresis mechanism was implemented. This improves classification stability without adding significant computational burden.

Fully Embedded Edge Processing:

All signal processing and localization decisions are performed locally on the ESP32. This eliminates the need for cloud-based processing, reduces latency,

minimizes communication overhead, and improves scalability for large indoor deployments.

Low Infrastructure Requirement:

The system is designed to operate with a limited number of BLE beacons, reducing deployment cost and maintenance complexity. This makes the architecture suitable for educational institutions, small commercial spaces, and resource-limited environments.

These design considerations ensure that the proposed system achieves a balance between performance, cost, scalability, and implementation simplicity.

CHAPTER 4

HARDWARE IMPLEMENTATION

4.1 HARDWARE OVERVIEW

The hardware implementation of the proposed zone-based BLE indoor localization system consists of an ESP32 development board functioning as the localization receiver and fixed BLE beacon modules deployed at known indoor locations. The hardware design focuses on low power consumption, simplicity of deployment, and reliable real-time signal acquisition.

The ESP32 platform serves as the central processing unit of the system. It performs continuous BLE scanning, RSSI acquisition, signal filtering, zone classification, and output generation. All computational tasks are executed locally without the need for external servers or cloud infrastructure.

BLE beacon modules act as fixed transmitters that periodically broadcast advertisement packets containing unique identifiers. These packets are received by the ESP32 and used for proximity-based zone determination.

The hardware configuration was selected to ensure minimal infrastructure requirement, low deployment cost, energy efficiency, and reliable real-time operation. The design prioritizes scalability and suitability for indoor environments such as classrooms, laboratories, hospital wards, and small commercial spaces.

4.2 ESP32 DEVKIT V1 (ESP32-WROOM-32) MODULE DESCRIPTION

The localization receiver is implemented using the ESP32 DevKit V1 development board based on the ESP32-WROOM-32 module. The ESP32 is

a low-cost, low-power System-on-Chip (SoC) developed by Espressif Systems, featuring integrated Wi-Fi and Bluetooth capabilities.

The ESP32-WROOM-32 module is built around a dual-core Tensilica Xtensa LX6 processor operating at up to 240 MHz. It includes 520 KB of SRAM and supports external flash memory for program storage. The integrated Bluetooth Low Energy (BLE 4.2) functionality enables continuous scanning of advertisement packets transmitted by nearby BLE beacon nodes.

Key hardware features relevant to this project include:

- Integrated 2.4 GHz Wi-Fi and Bluetooth Low Energy (BLE) radio
- Support for BLE scanning and advertisement packet parsing
- Dual-core 32-bit processor up to 240 MHz
- Low-power operating modes
- UART interface for serial debugging
- On-board 5V to 3.3V voltage regulation

In the proposed system, the ESP32 DevKit V1 functions as the central processing unit of the localization receiver. It continuously scans for BLE advertisement packets, extracts RSSI values, performs sliding window averaging, executes threshold-based zone classification, and generates real-time output.

The selection of the ESP32 DevKit V1 was motivated by its integrated BLE capability, sufficient processing performance, low cost, and suitability for embedded real-time indoor localization applications.

4.3 BLE BEACON MODULE DESCRIPTION

In the proposed system, BLE beacon functionality is implemented using an additional ESP32 DevKit V1 (ESP32-WROOM-32) development board configured in BLE advertisement mode. Instead of using a dedicated commercial beacon device, a programmable ESP32 platform was utilized to provide flexibility in configuring advertisement parameters such as transmission interval, advertisement data format, and broadcast power.

The beacon ESP32 operates in BLE advertising mode and periodically transmits advertisement packets containing a unique identifier. These packets are broadcast in the 2.4 GHz ISM band and are received by the localization receiver ESP32 operating in continuous scanning mode.

The use of an ESP32 as a beacon offers several advantages:

- Configurable advertisement interval
- Adjustable transmission power
- Customizable beacon identifier
- Easy firmware modification
- Cost-effective deployment

During experimental evaluation, the beacon was configured to transmit advertisement packets at fixed intervals to ensure consistent RSSI sampling. The transmission power was kept constant throughout the experiments to maintain stable signal characteristics.

4.4 CIRCUIT DIAGRAM AND CONNECTIONS

The hardware setup consists of two ESP32 DevKit V1 boards: one configured as a BLE beacon (transmitter) and the other operating as the localization receiver (scanner).

The localization receiver ESP32 is connected to a laptop via a USB cable. The USB connection serves two purposes: supplying 5V power to the board and enabling serial communication for real-time monitoring of RSSI values and zone classification output through the serial monitor.

The BLE beacon ESP32 is powered using an external regulated 5V power supply. The 5V supply is connected to the VIN (5V) pin and GND of the ESP32 DevKit board. The onboard voltage regulator converts the input voltage to 3.3V required by the ESP32-WROOM-32 module.

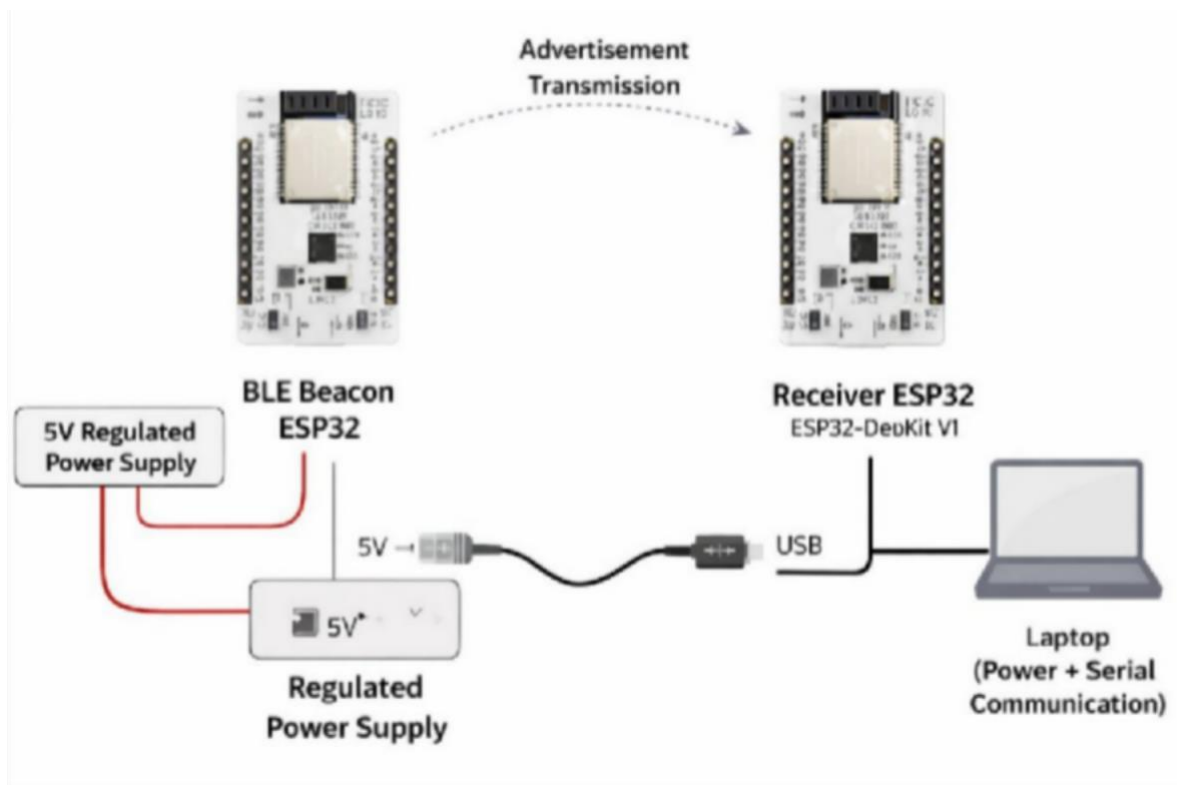


Figure 4.1 Circuit Diagram of the Proposed BLE-Based Indoor Localization System

The basic electrical connections are summarized below:

- Beacon ESP32:

- 5V → VIN pin
 - GND → GND pin
- Receiver ESP32:
- USB cable → Laptop (Power + Serial Communication)

4.5 HARDWARE PROTOTYPE SETUP

The experimental hardware prototype consists of two ESP32 DevKit V1 boards arranged in a controlled indoor environment. One ESP32 board operates in BLE advertisement mode and acts as a fixed beacon transmitter. The second ESP32 board functions as the localization receiver and is connected to a laptop for power supply and serial monitoring.

The BLE beacon ESP32 was positioned at a fixed location within the indoor test area. A regulated 5V power supply was used to ensure stable operation and consistent signal transmission. The advertisement interval and transmission power were kept constant throughout the experimental evaluation to maintain uniform signal conditions.

The receiver ESP32 was placed at predefined distances corresponding to Near, Mid, and Far zones. RSSI values were collected in real time using the serial monitor interface on the connected laptop. The indoor test area covered approximately 100 sq.ft, with a maximum separation distance of 3.25 meters between the beacon and receiver.

During experimentation, 90 RSSI samples were collected, comprising 30 samples per zone. A sliding window size of 8 samples was used for RSSI averaging, and the BLE scan duration was configured to 2 seconds per measurement cycle.

The prototype setup enabled real-time evaluation of zone classification accuracy, RSSI stability, and distance range mapping performance under practical indoor conditions.

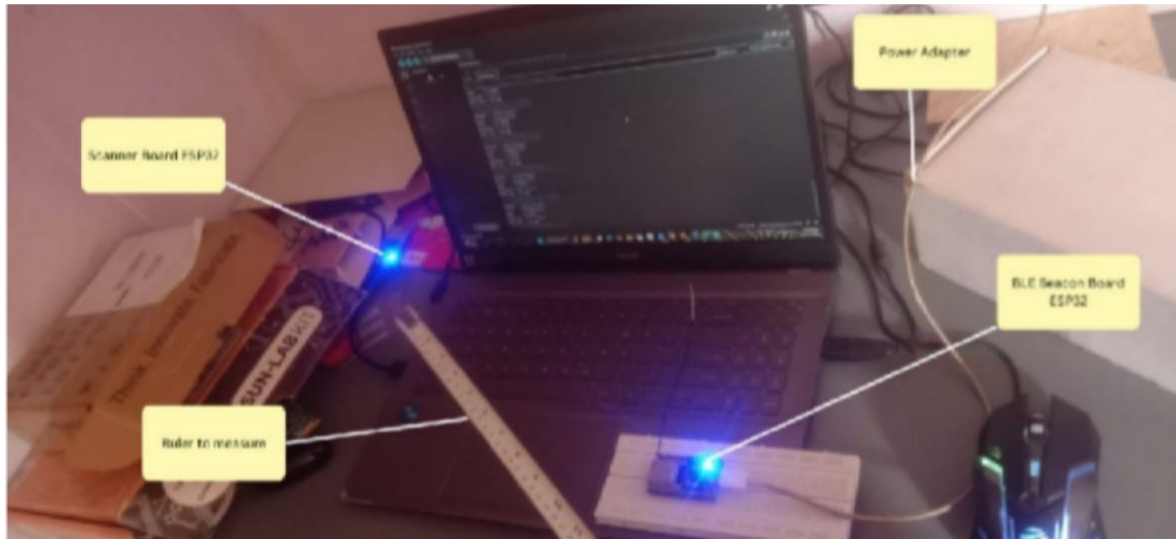


Figure 4.2 Hardware Prototype of the Proposed Zone-Based BLE Indoor Localization System

The hardware prototype shown in Figure 4.2 illustrates the complete experimental setup used for system validation. The ESP32 receiver board is connected to a laptop via USB for power supply and serial communication. The BLE beacon ESP32 board is powered using an external regulated 5V power adapter and is positioned at a fixed reference location. A measuring scale was used to accurately position the receiver board at predefined distances corresponding to Near, Mid, and Far zones. This setup enabled controlled indoor experimentation and reliable RSSI data collection.

CHAPTER 5

SOFTWARE IMPLEMENTATION

5.1 SOFTWARE DEVELOPMENT ENVIRONMENT

The software implementation of the proposed zone-based BLE indoor localization system was developed using the Arduino IDE environment with ESP32 board support package installed. The firmware was written in Embedded C/C++ and uploaded to the ESP32 DevKit V1 (ESP32-WROOM-32) development boards.

The BLE functionality was implemented using the ESP32 BLE library, which provides APIs for configuring BLE advertisement, scanning, and RSSI acquisition. Serial communication was used for debugging and monitoring real-time RSSI values and zone classification outputs through the serial monitor interface.

The receiver ESP32 was programmed in continuous BLE scanning mode, while the beacon ESP32 was configured in advertisement mode with fixed transmission intervals and predefined beacon identifiers.

The receiver ESP32 operates in continuous BLE scanning mode to detect advertisement packets transmitted by the beacon ESP32. The BLE scanner is initialized during system start up and configured with fixed scan interval and scan window duration.

Upon detecting a valid advertisement packet, the receiver extracts the Received Signal Strength Indicator (RSSI) value associated with the beacon.

Only packets matching the predefined beacon identifier are processed to ensure reliable localization.

The BLE scanning process follows these steps:

1. Initialize BLE device
2. Configure scanning parameters
3. Start continuous scan
4. Detect beacon advertisement
5. Extract RSSI value

A simplified code snippet for initializing BLE scanning is shown below:

```
BLEDevice::init("");  
  
BLEScan* pBLEScan = BLEDevice::getScan();  
  
pBLEScan->setActiveScan(true);  
  
pBLEScan->setInterval(100);  
  
pBLEScan->setWindow(99);  
  
pBLEScan->start(scanTime, false);
```

5.2 RSSI FILTERING IMPLEMENTATION

Due to multipath fading, shadowing, and environmental interference, raw RSSI values obtained from BLE advertisement packets exhibit significant short-term fluctuations. Direct usage of raw RSSI values results in unstable zone classification and frequent boundary oscillations.

To improve signal stability, a sliding window moving average filter was implemented directly on the ESP32. A fixed number of recent RSSI samples

are stored in a buffer, and the average of these samples is computed to obtain a filtered RSSI value.

The filtered RSSI is calculated using the following equation:

$$\bar{r} = \frac{1}{N} \sum_{i=1}^N r(i)$$

Where:

$r(i)$ = Instantaneous RSSI sample

N = Window size

$\bar{r}$ = Averaged RSSI value

In the experimental setup, the window size was fixed at **N = 8 samples**, providing a balance between noise reduction and system responsiveness.

The filtering process follows these steps:

1. Store new RSSI sample in buffer
2. Update buffer index
3. Compute sum of stored samples
4. Calculate average RSSI
5. Use filtered RSSI for zone decision logic

A simplified implementation snippet is shown below:

```
#define WINDOW_SIZE 8

int rssiBuffer[WINDOW_SIZE];

int index = 0;

long sum = 0;
```

```

void updateRSSI(int newRSSI)

{ sum -= rssiBuffer[index];

  rssiBuffer[index] = newRSSI;

  sum += newRSSI;

  index = (index + 1) % WINDOW_SIZE;

}

```

```

float getFilteredRSSI() {

  return (float)sum / WINDOW_SIZE;

}

```

The lightweight moving average approach was selected instead of computationally intensive filtering techniques such as Kalman filtering to ensure real-time execution on the embedded ESP32 platform with minimal memory and processing overhead.

5.3 ZONE CLASSIFICATION LOGIC

The zone classification module determines the user's proximity zone based on the filtered RSSI value obtained from the sliding window averaging stage. The classification logic compares the filtered RSSI value with predefined threshold ranges corresponding to Near, Mid, and Far zones.

In the initial implementation phase, fixed threshold values were hardcoded based on experimental observation of RSSI ranges within the indoor test environment. These thresholds were determined empirically by measuring RSSI values at known distances.

However, during experimental evaluation, minor adjustments were made dynamically to refine zone boundaries and improve classification stability.

This hybrid approach enabled flexibility in adapting to environmental variations without significantly increasing computational complexity.

A typical threshold configuration used in the experimental setup is shown below:

- Near Zone (Zone A): $\text{RSSI} > -40 \text{ dBm}$
- Mid Zone (Zone B): $-55 \text{ dBm} < \text{RSSI} \leq -40 \text{ dBm}$
- Far Zone (Zone C): $\text{RSSI} \leq -55 \text{ dBm}$

The classification decision is implemented using conditional logic as follows:

```
String determineZone(float filteredRSSI)
```

```
    { if (filteredRSSI > -40) {  
        return "Zone A (Near)";  
    }  
    else if (filteredRSSI > -55)  
        { return "Zone B (Mid)";  
        }  
    else {  
        return "Zone C (Far)";  
    }  
}
```

The threshold-based zone decision approach was selected instead of coordinate-based trilateration to reduce computational overhead and enhance robustness against RSSI instability.

5.4 HYSTERESIS-BASED ZONE STABILIZATION

Due to minor RSSI fluctuations near threshold boundaries, frequent transitions between adjacent zones may occur if classification decisions are

made instantaneously. To prevent unstable zone switching, a hysteresis-based stabilization mechanism was implemented.

In the proposed system, a new zone is accepted only if it is detected consistently for **three consecutive measurement cycles**. If the zone classification changes but does not persist for three cycles, the previous zone is retained.

This approach significantly reduces oscillation near boundary regions and improves overall classification stability.

The hysteresis logic operates as follows:

1. Compute filtered RSSI
2. Determine tentative zone
3. Compare with previously confirmed zone
4. If different, increment transition counter
5. Accept new zone only if counter ≥ 3
6. Otherwise retain previous zone

A simplified implementation is shown below:

```
String currentZone = "";
```

```
String previousZone = "";
```

```
int stabilityCounter = 0;
```

```
String applyHysteresis(String detectedZone) {
```

```
    if (detectedZone == previousZone)
```

```
        { stabilityCounter++;
```

```

    } else {

        stabilityCounter = 1;

        previousZone = detectedZone;

    }

    if (stabilityCounter >= 3)

        { currentZone = detectedZone;

        }

    return currentZone;

}

```

The hysteresis mechanism ensures that short-term RSSI disturbances do not cause unnecessary zone transitions. This improves user experience and enhances reliability in practical indoor environments.

5.5 DISTANCE RANGE ESTIMATION IMPLEMENTATION

In addition to zone classification, the system estimates an approximate distance range using the log-distance path loss model. The purpose of this estimation is not to compute precise spatial coordinates, but to provide contextual proximity information corresponding to predefined zones.

The distance is estimated using the following equation:

$$\square = \square_0 \times 10^{\frac{\square_0 \times \bar{\square}}{10 \square}}$$

where:

$\square$ = Estimated distance between beacon and receiver

$\square_0$ = Reference distance (1 meter)

P_{00} = RSSI value measured at reference distance

$\overline{P}$ = Filtered RSSI value

n = Indoor path loss exponent

During experimentation, the reference RSSI value P_{00} was measured at a distance of 1 meter from the beacon. The path loss exponent n was assumed as a typical indoor constant value to approximate signal attenuation under normal indoor conditions.

The estimated distance value was mapped to predefined proximity ranges corresponding to Near, Mid, and Far zones rather than being used for coordinate-level localization.

A simplified implementation snippet is shown below:

```
float calculateDistance(float filteredRSSI) {  
  
    float d0 = 1.0;    // reference distance (1 meter)  
  
    float P0 = -45.0;  // measured RSSI at 1 meter  
  
    float n = 2.5;     // assumed indoor path loss exponent  
  
    float exponent = (P0 - filteredRSSI) / (10 * n);  
  
    float distance = d0 * pow(10, exponent);  
  
    return distance;  
}
```

The use of a simplified path loss model ensures low computational complexity and real-time execution on the ESP32 platform.

CHAPTER 6

RESULTS AND ANALYSIS

6.1 EXPERIMENTAL CONDITIONS

The experimental evaluation of the proposed zone-based BLE indoor localization system was conducted in a controlled indoor environment covering approximately 100 sq.ft ($\approx 9.29 \text{ m}^2$). The BLE beacon ESP32 was positioned at a fixed reference location, while the receiver ESP32 was placed at predefined distances corresponding to Near, Mid, and Far zones in this indoor positioning system project.

The maximum separation distance between the beacon and receiver was 3.25 meters. RSSI samples were collected in static conditions to minimize motion-induced signal variations. A total of 90 RSSI samples were recorded, comprising 30 samples per zone.

The BLE scan duration was configured to 2 seconds per cycle, and a sliding window size of 8 samples was used for RSSI filtering. All measurements were recorded using the serial monitoring interface connected to the receiver ESP32.

6.2 ZONE-WISE RSSI RANGE ANALYSIS

The measured RSSI values were categorized into three predefined proximity zones based on experimentally observed signal ranges. The average RSSI ranges and corresponding distance intervals are summarized in the given Table 6.1.

Zone	RSSI Range (dBm)	Distance Range (m)
Zone A (Near)	-26 to -39	< 0.5
Zone B (Mid)	-48 to -55	0.5 – 1.5
Zone C (Far)	-56 to -62	1.5 – 3.25

Table 6.1 Zone Distance Ranges and Corresponding RSSI Values

6.3 ZONE DETECTION ACCURACY

To evaluate system performance, zone classification accuracy was calculated for both unfiltered (raw RSSI) and filtered (sliding window averaged) implementations.

Zone	Samples	Accuracy (unfiltered)	Accuracy (Filtered)
Zone A (Near)	30	78%	93%
Zone B (Mid)	30	74%	91%
Zone C (Far)	30	71%	89%

Table 6.2 presents the comparison of classification accuracy for each zone

6.4 RSSI STABILITY ANALYSIS

Raw RSSI values obtained from BLE advertisement packets exhibited significant short-term fluctuations due to multipath fading, environmental reflections, and minor positional variations. These fluctuations caused instability in direct zone classification when raw RSSI values were used.

To evaluate signal stability improvement, both raw RSSI and filtered RSSI values were monitored and compared. The sliding window moving average filter significantly reduced short-term variance and smoothed abrupt signal spikes.

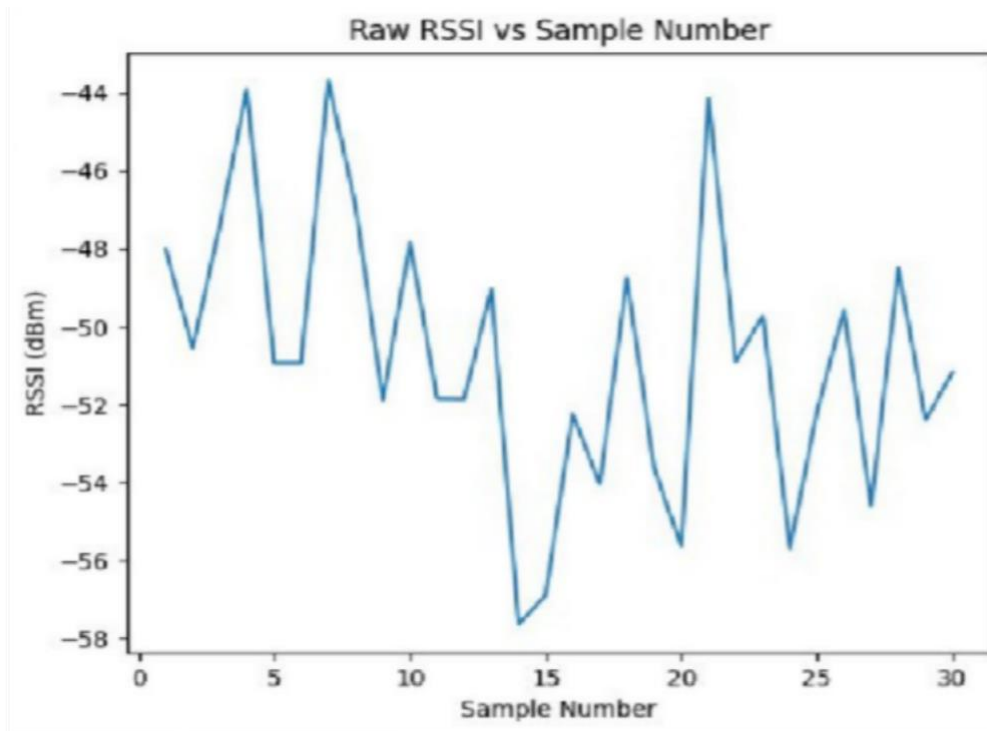


Figure 6.1 Raw RSSI Variations across Measurement Samples

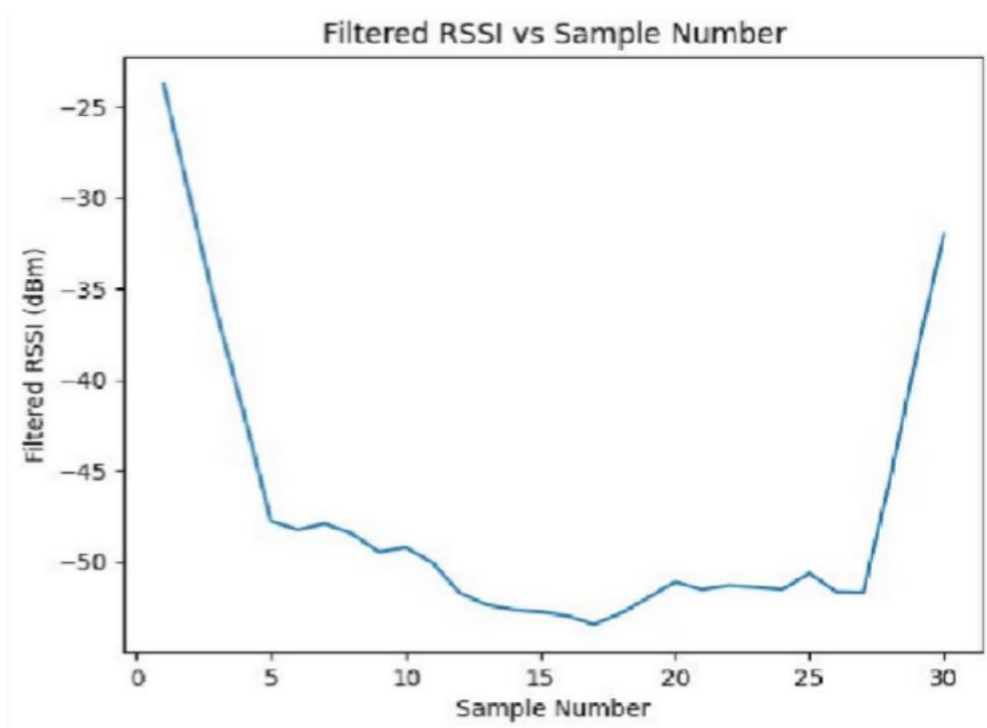


Figure 6.2 Filtered RSSI Variations after Sliding Window Averaging

The filtered RSSI values demonstrated reduced oscillation near threshold boundaries, thereby minimizing unnecessary zone transitions. When combined with the hysteresis-based stabilization mechanism, the system achieved consistent zone classification even when the receiver was positioned close to zone edges.

The improvement in stability directly contributed to the observed increase in classification accuracy from approximately 74% (unfiltered) to 91% (filtered implementation). This confirms that lightweight RSSI filtering is effective in enhancing zone-level indoor localization performance without introducing significant computational complexity.

From Figures 6.1 and 6.2, the graphical comparison clearly demonstrates that sliding window averaging reduces short-term signal oscillations and stabilizes RSSI behaviour, thereby improving zone classification reliability.

6.5 OUTPUT VALIDATION AND DISCUSSION

The final system output consists of the identified proximity zone, estimated distance range, filtered RSSI value, and contextual information such as floor and product identifier. The output was monitored in real time through the serial interface connected to the receiver ESP32.

To validate correctness, the physical distance between the beacon and receiver was measured using a ruler and compared with the predicted zone classification. The system consistently classified the receiver into the correct zone based on predefined distance boundaries.

The given below table which represents output samples obtained during the experiments.

Product Name	Floor	Zone	Distance Range (m)	RSSI Avg (dBm)
Dairy Product	Ground Floor	Zone C (Far)	> 1.5	-62
Dairy Product	Ground Floor	Zone B (Mid)	0.5 – 1.5	-48
Dairy Product	Ground Floor	Zone A (Near)	<0.5	-38

Table 6.3. Representative output samples obtained during experimentation are summarized.

The experimental results confirm that the proposed zone-based BLE localization system provides reliable proximity classification without estimating precise spatial coordinates. The combination of sliding window RSSI filtering and hysteresis-based stabilization significantly reduced zone oscillation near threshold boundaries. The system achieved an overall classification accuracy of approximately 91%, demonstrating that lightweight embedded processing on the ESP32 platform is sufficient for practical indoor zone-level localization applications.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

This project presented the design and implementation of a zone-based BLE indoor localization system using the ESP32 DevKit V1 (ESP32-WROOM-32) platform. The proposed system focuses on stable proximity-based zone classification rather than precise coordinate estimation, thereby reducing computational complexity and improving robustness in dynamic indoor environments.

The system integrates BLE scanning, sliding window RSSI filtering, threshold-based zone classification, hysteresis stabilization, and distance range estimation within a fully embedded architecture. All processing operations are executed locally on the ESP32 without reliance on cloud-based computation.

Experimental evaluation conducted in a controlled indoor environment demonstrated that RSSI filtering significantly improved signal stability and reduced zone oscillation near decision boundaries. The classification accuracy improved from approximately 74% (raw RSSI) to 91% (filtered implementation), confirming the effectiveness of lightweight embedded signal conditioning.

The proposed architecture achieves low hardware complexity, low deployment cost, and reliable real-time performance, making it suitable for practical indoor applications such as asset tracking, aisle-level navigation, and context-aware services.

7.2 FUTURE WORK

Although the proposed system demonstrates reliable zone-level localization, several improvements can be considered for future development:

- Deployment of multiple beacon nodes for multi-room or multi-floor environments
- Adaptive threshold tuning based on environmental calibration
- Integration of advanced filtering techniques such as Kalman filtering for enhanced accuracy
- Mobile application interface for real-time user interaction
- Battery-powered beacon optimization for long-term deployment
- Hybrid localization combining BLE with additional sensors such as IMU or Wi-Fi

Future work may also explore coordinate-level estimation using trilateration while maintaining low computational overhead.

APPENDIX A

RECEIVER ESP32 SOURCE CODE

ESP32 BLE SCANNER – ZONE BASED INDOOR POSITIONING SYSTEM

Board : ESP32 DevKit V1 (ESP32-WROOM-32)

Role : Localization Receiver (Scanner)

/******

```
#include <BLEDevice.h>

#include <BLEScan.h>

#include <BLEAdvertisedDevice.h>

#include <math.h>

#define PRODUCT_NAME "Dairy Product"

#define FLOOR_NAME "Ground Floor"

#define TARGET_MAC "e0:8c:fe:5b:1b:66"

#define NEAR_TH -40

#define MID_TH -55

#define RSSI_WINDOW 8

#define SCAN_TIME 2

#define STABILITY_LIMIT 3

#define REF_DISTANCE 1.0

#define REF_RSSI -45.0

#define PATH_LOSS_EXP 2.5

enum Zone { NEAR, MID, FAR };

BLEScan* pBLEScan;

int rssiBuf[RSSI_WINDOW];

int idx = 0;

bool filled = false;
```

```

Zone lockedZone = FAR;

Zone previousZone = FAR;

int stabilityCounter = 0;

class ScanCallbacks : public BLEAdvertisedDeviceCallbacks
{
    void onResult(BLEAdvertisedDevice d) {
        std::string mac = d.getAddress().toString();

        if (mac == TARGET_MAC) {
            int rssi = d.getRSSI();

            rssiBuf[idx++] = rssi;

            if (idx >= RSSI_WINDOW) {
                idx = 0;

                filled = true;
            }
        }
    }
};

int avgRSSI()
{
    int sum = 0;

    int count = filled ? RSSI_WINDOW : idx;

    if (count == 0) return -100;

    for (int i = 0; i < count; i++)
        { sum += rssiBuf[i];
        }

    return sum / count;
}

Zone decideZone(int rssi) {
    if (rssi >= NEAR_TH) return NEAR;

    if (rssi >= MID_TH) return MID;

    return FAR;
}

```

```

}

float calculateDistance(float filteredRSSI) {
    float exponent = (REF_RSSI - filteredRSSI) / (10 * PATH_LOSS_EXP);
    float distance = REF_DISTANCE * pow(10, exponent);
    return distance;
}

const char* zoneStr(Zone z) {
    if (z == NEAR) return "Zone A (Near)";
    if (z == MID) return "Zone B (Mid)";
    return "Zone C (Far)";
}

const char* distRangeStr(Zone z)
{
    if (z == NEAR) return "< 0.5
m";
    if (z == MID) return "0.5 – 1.5 m";
    return "> 1.5 m";
}

void printOutput(int rssi, float distance)
{
    Serial.println(" ----- ");
    Serial.print("Product : "); Serial.println(PRODUCT_NAME);
    Serial.print("Floor : "); Serial.println(FLOOR_NAME);
    Serial.print("Zone  : "); Serial.println(zoneStr(lockedZone));
    Serial.print("Distance : "); Serial.println(distRangeStr(lockedZone));
    Serial.print("RSSI avg : "); Serial.print(rssi); Serial.println(" dBm");
    Serial.print("Est. d  : "); Serial.print(distance); Serial.println(" m");
    Serial.println(" ----- ");
    Serial.print("LOG,");
    Serial.print(millis());

```

```
Serial.print(",");
```

```

Serial.print(rssi);

Serial.print(",");

Serial.print(distance);

Serial.print(",");

Serial.println(zoneStr(lockedZone));
}

void setup()
{
  Serial.begin(115200);

  delay(1000);

  Serial.println("ESP32 BLE ZONE SCANNER STARTED");

  BLEDevice::init("");

  pBLEScan = BLEDevice::getScan();

  pBLEScan->setAdvertisedDeviceCallbacks(new ScanCallbacks());

  pBLEScan->setActiveScan(true);
}

void loop() {
  pBLEScan->start(SCAN_TIME, false);

  pBLEScan->clearResults();

  if (!filled && idx < RSSI_WINDOW)
  {
    Serial.println("Collecting RSSI samples...");

    delay(500);

    return;
  }

  int filteredRSSI = avgRSSI();

  Zone detectedZone = decideZone(filteredRSSI);

  if (detectedZone == previousZone)
  {
    stabilityCounter++;
  } else
  {
    stabilityCounter =

```


1;

```
    previousZone = detectedZone;
}

if (stabilityCounter >= STABILITY_LIMIT)
    { lockedZone = detectedZone;
    }
float distance = calculateDistance(filteredRSSI);
printOutput(filteredRSSI, distance);
delay(1000);
}
```

APPENDIX B

BEACON ESP32 SOURCE CODE

ESP32 BLE BEACON – ZONE BASED INDOOR POSITIONING SYSTEM

Board : ESP32 DevKit V1 (ESP32-WROOM-32)

Role : BLE Advertisement Beacon

/*****
 *****/

```
#include <BLEDevice.h>
```

```
#include <BLEUtils.h>
```

```
#include <BLEServer.h>
```

```
#include <BLEAdvertising.h>
```

```
#define BEACON_NAME "ZoneBeacon"
```

```
#define TX_POWER    ESP_PWR_LVL_P9
```

```
#define ADV_INTERVAL 100
```

BLEAdvertising *pAdvertising;

```
void setup()
```

```
{ Serial.begin(115200);
```

```
delay(1000);
```

```
Serial.println("ESP32 BLE BEACON STARTED");
```

```
BLEDevice::init(BEACON_NAME);
```

```
BLEServer *pServer = BLEDevice::createServer();
```

```
pAdvertising = BLEDevice::getAdvertising();
```

```
pAdvertising->setScanResponse(false);
```

```
pAdvertising->setMinInterval(ADV_INTERVAL);
```

```
pAdvertising->setMaxInterval(ADV_INTERVAL);
```

```
pAdvertising->setTxPower(TX_POWER);
```

```
pAdvertising->start();
```

```
    Serial.println("Beacon advertising started...");  
}  
void loop()  
{ delay(2000);  
}
```

REFERENCES

- [1] M. Alshami, H. Alshammari, and M. Younis, “Low-cost BLE-based indoor localization using adaptive RSSI filtering,” *IEEE Access*, vol. 12, pp. 18421–18435, 2024.
- [2] J. Kim and S. Lee, “Zone-based indoor positioning using BLE beacons and lightweight signal processing,” *Sensors*, vol. 24, no. 3, pp. 1158–1172, 2024.
- [3] A. K. Paul and T. Sato, “Bluetooth Low Energy-based indoor positioning: A practical implementation study,” *IEEE Internet of Things Journal*, vol. 11, no. 2, pp. 2567–2578, 2024.
- [4] R. Singh, P. Sharma, and V. Gupta, “RSSI stabilization for indoor localization using moving average and hysteresis techniques,” *IEEE Access*, vol. 11, pp. 98234–98245, 2023.
- [5] L. Chen, Y. Wang, and H. Zhang, “Edge-based BLE indoor positioning with reduced computational complexity,” *IEEE Transactions on Consumer Electronics*, vol. 69, no. 4, pp. 455–463, 2023.
- [6] S. Park and K. Cho, “Performance evaluation of BLE RSSI filtering techniques for indoor zone classification,” *Sensors*, vol. 23, no. 12, pp. 5621–5635, 2023.
- [7] Y. Zafari, A. Gkelias, and K. K. Leung, “A survey of indoor localization systems and technologies,” *IEEE Communications Surveys & Tutorials*, vol. 23, no. 4, pp. 2568–2599, 2023.

- [8] Espressif Systems, *ESP32-WROOM-32 Datasheet*, 2023.
- [9] Espressif Systems, *ESP32 Technical Reference Manual*, 2023.
- [10] H. Liu, H. Darabi, P. Banerjee, and J. Liu, “Survey of wireless indoor positioning techniques and systems,” *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 37, no. 6, pp. 1067–1080, 2022.
- [11] S. He and S.-H. G. Chan, “Wi-Fi fingerprint-based indoor positioning: Recent advances and comparisons,” *IEEE Communications Surveys & Tutorials*, vol. 18, no. 1, pp. 466–490, 2021.
- [12] R. Faragher and R. Harle, “Location fingerprinting with Bluetooth Low Energy beacons,” *IEEE Journal on Selected Areas in Communications*, vol. 33, no. 11, pp. 2418–2428, 2020.
- [13] S. Subedi and J.-Y. Pyun, “Practical fingerprinting localization for indoor positioning using BLE beacons,” *IEEE Access*, vol. 8, pp. 123–134, 2020.
- [14] X. Wang, L. Gao, S. Mao, and S. Pandey, “CSI-based fingerprinting for indoor localization: A deep learning approach,” *IEEE Transactions on Vehicular Technology*, vol. 66, no. 1, pp. 763–776, 2019.
- [15] P. Davidson and R. Piché, “A survey of selected indoor positioning methods for smartphones,” *IEEE Communications Surveys & Tutorials*, vol. 19, no. 2, pp. 1347–1370, 2018.

PO & PSO ATTAINMENT

PO. No.	Graduate Attribute	Attained	Justification
PO 1	Engineering Knowledge	Yes	The project applies wireless communication principles, RSSI-based distance modeling, and embedded system design using ESP32 and BLE. Core concepts of signal propagation and path loss were implemented for indoor localization.
PO 2	Problem Analysis	Yes	The system analyzes indoor localization challenges such as RSSI fluctuations, multipath effects, and signal attenuation. Filtering and hysteresis techniques were applied to improve stability and accuracy.
PO 3	Design/Development of Solutions	Yes	A zone-based BLE localization architecture was designed using ESP32 beacon and scanner nodes. Hardware integration, firmware development, and threshold-based

			classification were implemented..
PO 4	Conduct Investigations of Complex Problems	Yes	Experimental testing was performed at different distances to analyze RSSI variation and classification accuracy. Data analysis validated the effectiveness of filtering techniques.
PO 5	Modern Tool Usage	Yes	Arduino IDE, ESP32 BLE libraries, and serial monitoring tools were used for development and testing. Embedded C/C++ programming enabled real-time implementation.
PO 6	The Engineer and Society	Yes	The system supports applications such as asset tracking and smart building navigation. It enhances operational efficiency in indoor environments.
PO 7	Environment and Sustainability	Yes	Low-power BLE technology and minimal hardware infrastructure reduce energy consumption. The design supports sustainable indoor smart systems.

PO 8	Ethics	Yes	The system focuses on zone-level detection to reduce privacy concerns. Data was collected and used strictly for academic purposes.
PO 9	Individual and Team Work	Yes	The project involved firmware development, system integration, and testing. Collaborative efforts improved implementation and documentation quality.
PO10	Communication	Yes	The project is documented with structured reports, diagrams, graphs, and code explanations. Technical results are clearly presented.
PO11	Project Management and Finance	Yes	Cost-effective ESP32 modules and limited hardware components were used. The project was completed within budget and time constraints.

PO12	Life-long Learning	Yes	The project required learning emerging technologies such as BLE communication, embedded signal processing, and indoor localization algorithms. Continuous experimentation and problem-solving enhanced technical adaptability beyond regular coursework.
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